

Python Setup for ML

1. Recommended Python Version

As of 2026: Python 3.11

It has excellent support across:

- Scikit-learn
- XGBoost
- LightGBM
- PyTorch
- TensorFlow
- MLflow
- Airflow

2. Core Industry ML Package Setup

Install:

```
pip install numpy pandas scipy matplotlib seaborn
```

These are the basic data science libraries.

3. Machine Learning Libraries

Scikit-Learn

```
pip install scikit-learn
```

Used for:

- Classification
- Regression
- Clustering
- Feature Engineering
- Pipelines

XGBoost

```
pip install xgboost
```

Most common structured/tabular data model.

Examples:

- Churn prediction
- Fraud detection
- Credit scoring

LightGBM

```
pip install lightgbm
```

Widely used in:

- Fintech
- E-commerce
- Forecasting

4. Deep Learning

PyTorch

```
pip install torch torchvision torchaudio
```

Current industry favorite.

Applications:

- NLP
- Computer Vision
- Generative AI

Many AI teams prefer Meta's PyTorch ecosystem.

TensorFlow

```
pip install tensorflow
```

Still common in some enterprise environments.

5. Data Validation

```
pip install pandera
```

Example:

```
import pandera as pa
```

Used to validate:

- Data types
- Null values
- Business rules

Important in production pipelines.

6. Experiment Tracking

MLflow

```
pip install mlflow
```

Used for:

- Model tracking
- Experiment tracking
- Model registry

Many organizations use Databricks MLflow.

7. API Deployment

FastAPI

```
pip install fastapi uvicorn
```

Production inference example:

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/")
def health():
    return {"status": "ok"}
```

Run:

```
uvicorn app:app --reload
```

Very common for ML model serving.

8. Database Connectivity

PostgreSQL

```
pip install sqlalchemy psycopg2-binary
```

Example:

```
from sqlalchemy import create_engine

engine = create_engine(
    "postgresql://user:password@localhost/db"
)
```

Production ML rarely reads CSV files directly. Most data comes from databases or data warehouses.

Typical Industry Stack by Role

Role	Main Tools
Data Analyst	Pandas, SQL, Power BI/Tableau
Data Scientist	Pandas, Scikit-learn, XGBoost, MLflow
ML Engineer	PyTorch, FastAPI, Docker, Kubernetes
MLOps Engineer	MLflow, Airflow, Docker, Kubernetes
GenAI Engineer	PyTorch, Transformers, Vector DBs, RAG